Seungwon Yum

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Education

Seoul National University Seoul, South Korea
MS in Statistics Sep. 2025 – Present
Advisor: Professor Hee-Seok Oh

Ewha Womans University Seoul, South Korea
BS in Statistics, Minor in Computer Science Mar. 2021 – Aug. 2025
GPA: 4.08/4.3, Summa Cum Laude

Willamette University Oregon, United States
Exchange Program, Data Science Aug. 2023 – Dec. 2023

Research Interests

Statistical Analysis of Network Data; High-Dimensional Statistics; Dimension Reduction and Variable Selection; Multivariate Time Series Analysis; Missing Data

Research Experience

Multiscale Methods in Statistics Lab, Seoul National University
Graduate Researcher (Advisor: Professor Hee-Seok Oh) Dec. 2025 – Present

- Developing sparse principal component analysis in the graph frequency domain for interpretable dimension reduction and frequency-specific variable selection.
- Evaluating spectral power retention and reconstruction accuracy through simulations and applications to transportation and power networks.

Research Funding

2026 Master's Student Research Grant Sep. 2026 – Aug. 2027
Science and Engineering, National Research Foundation of Korea (NRF)

- Project: *Missing Value Analysis in Network Time Series Using Graph-Based Latent Factor Models*
- Principal Investigator; Total funding: KRW 12,000,000 (Grant No. RS-2026-25560499).

Honors & Awards

Grand Award, 4th Retail Data Utilization Competition Nov. 2025
Awarded by the Minister of Trade, Industry and Energy.

Excellence Award, MixUP AI Datathon May 2025
AI datathon in collaboration with WIZCORE.

W4 Scholar, Full Scholarship for Academic Excellence 2021 – 2024
Ewha Womans University

Teaching Experience

Seoul National University Seoul, South Korea
Teaching Assistant, Statistics Lab. Fall 2025

- Graded assignments and exams and answered undergraduate students' questions.

Industry Experience

Woori Bank, Big Data Platform Division Seoul, South Korea
Intern Jan. 2025 – Feb. 2025

- Analyzed CDP digital logs to inform marketing and customer retention strategies.
- Identified customer drop-off patterns across product application stages.

Projects

4th Retail Data Utilization Competition, Demand Forecasting Division
Team Leader, Presenter; Grand Award Sep. 2025 – Nov. 2025

- Developed a risk-weighted weekly demand forecasting framework for fulfillment centers.
- Designed a custom asymmetric sMAPE metric reflecting inventory cost asymmetry.
- Reduced overstock risk by approximately 11.5% through data-driven forecasting.

GNN-Based Skincare Recommendation System
Participant Oct. 2025 – Jan. 2026

- Developed a skincare recommendation system using graph-structured product, ingredient, and skin-type data.
- Implemented and compared GNN models including LightGCN and GraphSAGE.
- Constructed recommendation pipelines using ingredient similarity and review-based features.

MixUP AI Datathon
Presenter; Excellence Award May 2025

- Developed models for OK/NG steel quality prediction using boosting-based approaches.
- Improved model robustness through tension normalization and feature engineering.

2024 Weather Big Data Contest
Team Leader May 2024 – Jun. 2024

- Analyzed weather-fire correlations using meteorological and geospatial data.
- Handled missing values with interpolation and distance-based methods.

Skills & Languages

Software Python, R, SQL, SAS, PyTorch, LaTeX
Languages Korean (native), English (advanced)
Certifications ADsP, SQLD, Big Data Analysis Engineer